Time Series Analysis of Gold ETF Prices

Using Base Models, ARMA, ARIMA, and Box-Jenkins Models

Submitted by:
Name: Mohammed Hussain Nathani
GWID: G36308827
Course: DATS_6313_10

Project Overview

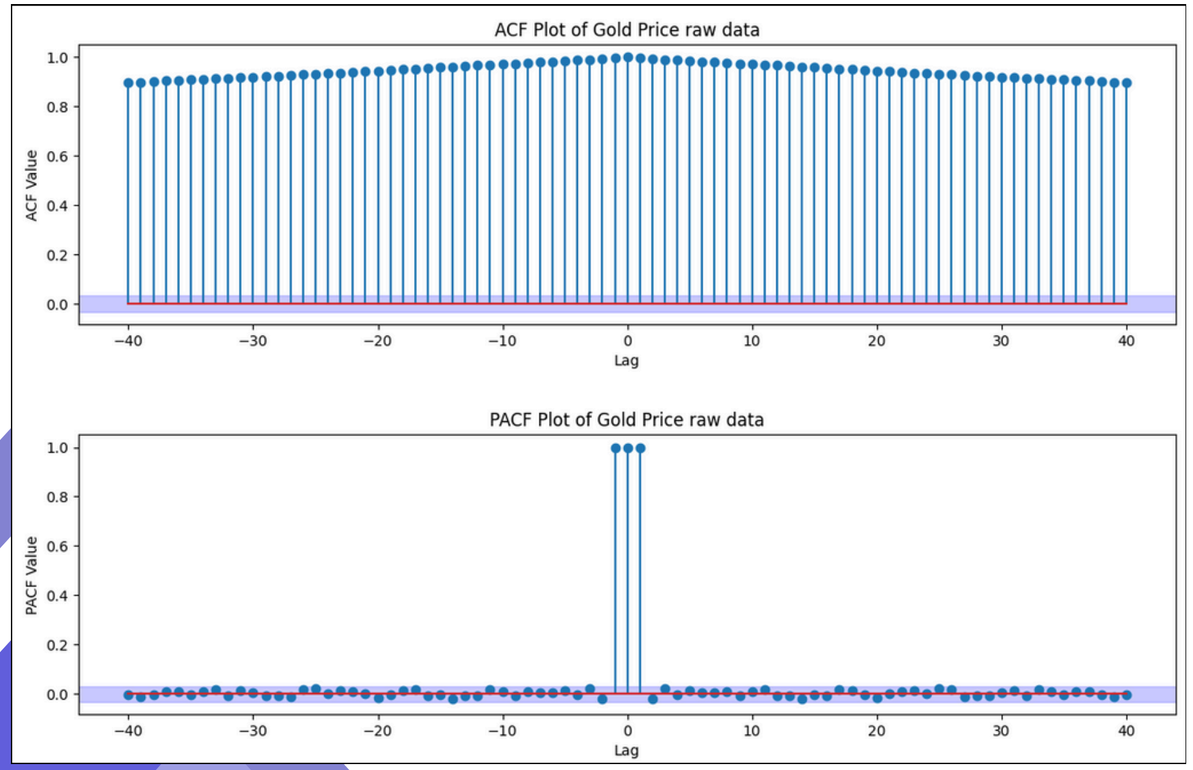
- Goal: Forecast Gold ETF Close Prices, aiming to uncover temporal patterns and develop predictive models for future price forecasting
- Dataset: 12 financial indicators (gold, silver, oil, GDP, CPI, etc.)
- Methods: Base Models, MLR, Holt-Winters, ARMA, ARIMA, Box-Jenkins

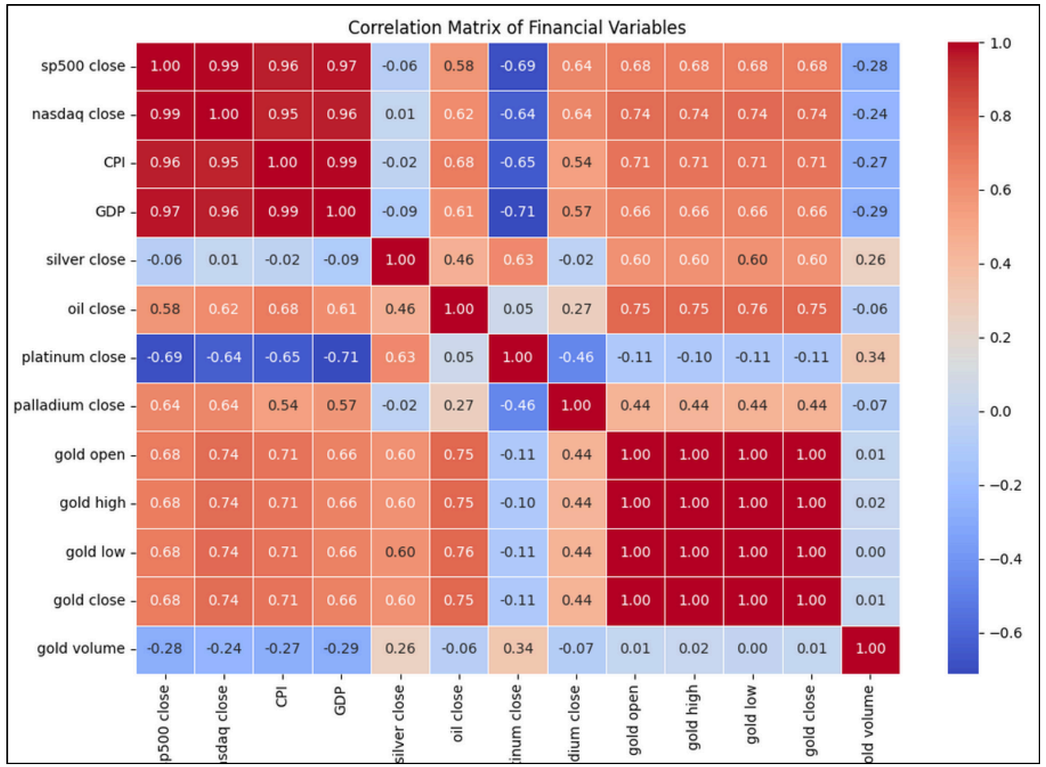
Data Summary + Trends

- Line chart of gold close price
- Mention: non-stationary pattern, visual trends

ACF/PACF and Correlation Insights

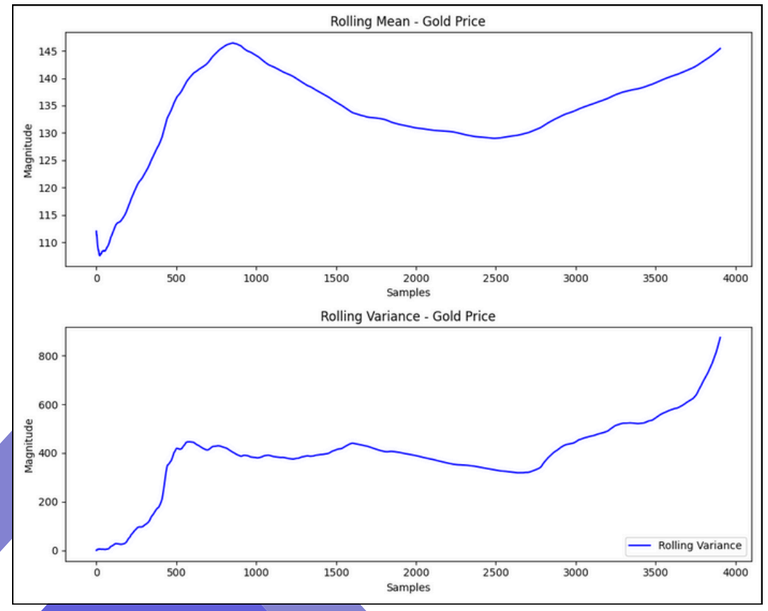
- Heatmap: feature relationships
- ACF: Decays slowly. PACF: Cut-Off

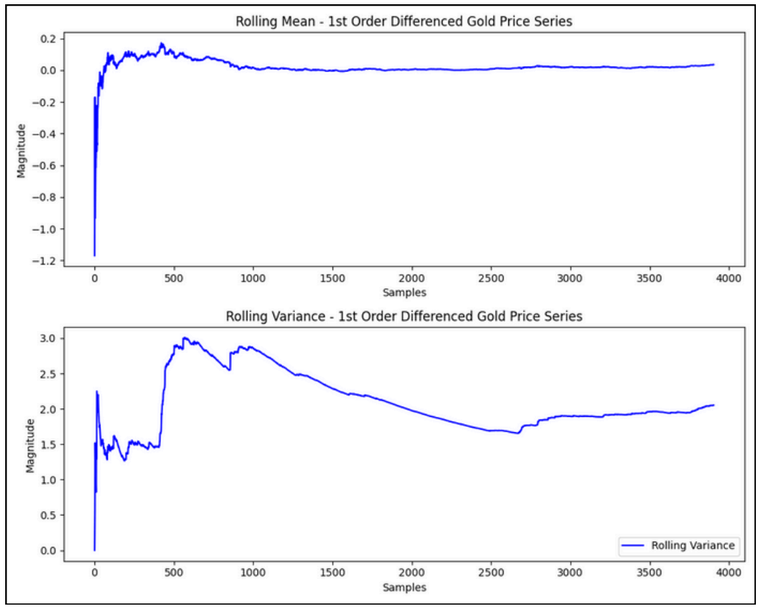


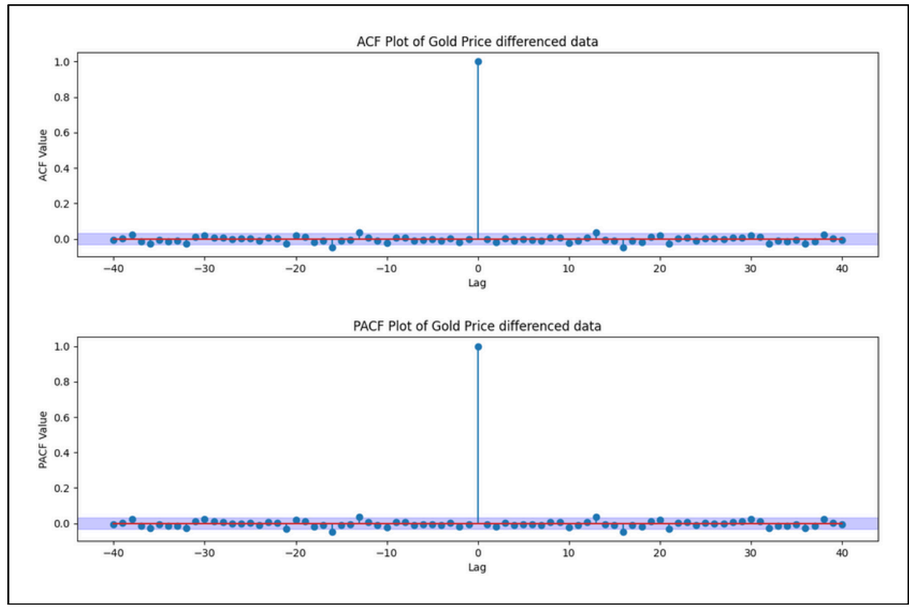


Stationarity Checks

- The Gold Close data was non-stationary. 1st order differencing performed.





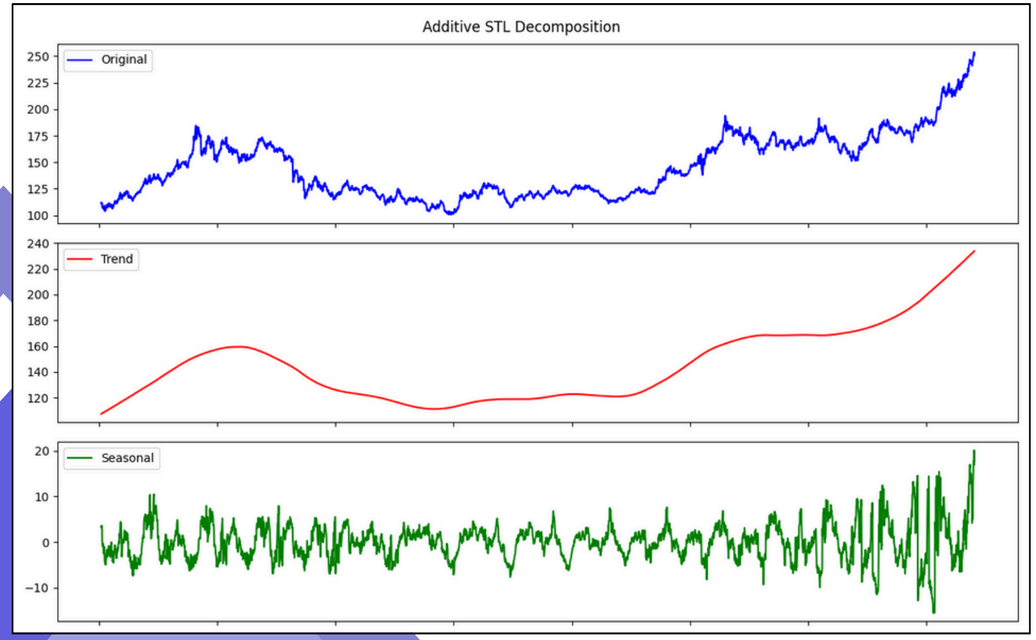


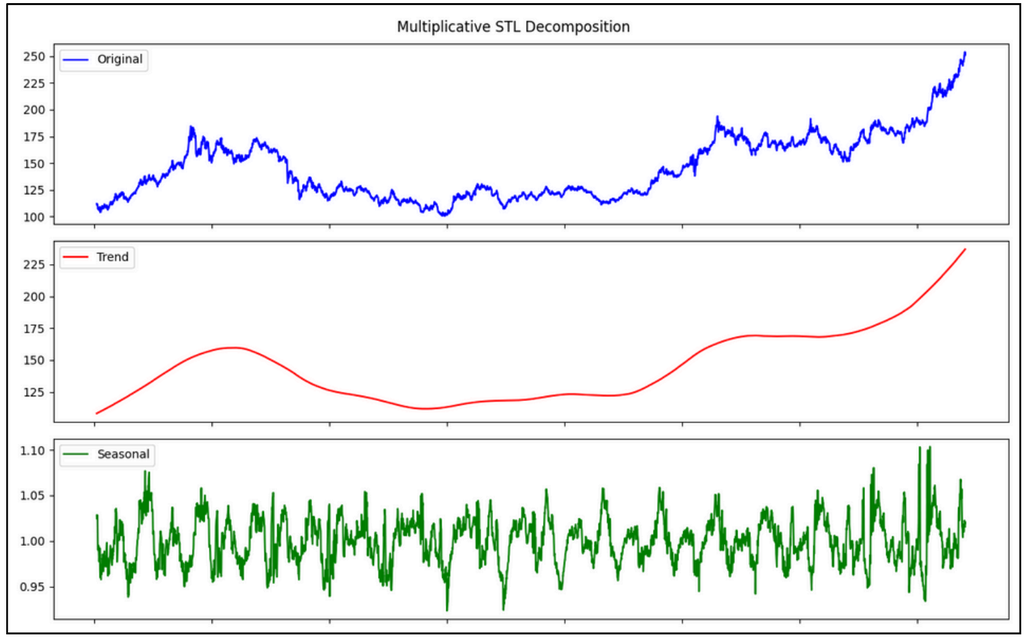
Decomposition Analysis

- Additive vs Multiplicative STL plots
- Strong trend, weak seasonality

- Additive Decomposition:
  Strength of Trend: 0.9406
  Strength of Seasonality: 0.0580

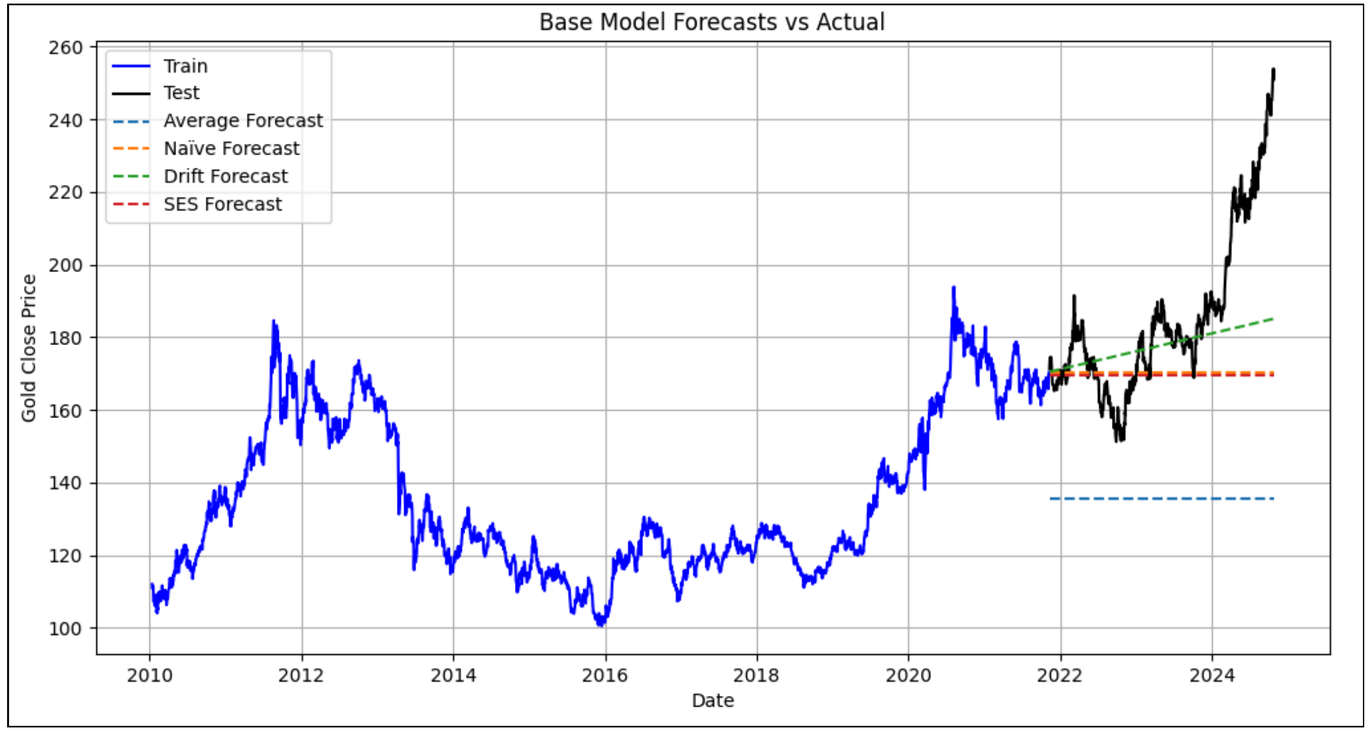
- Multiplicative Decomposition:
  Strength of Trend: 1.0000
  Strength of Seasonality: 0.1202





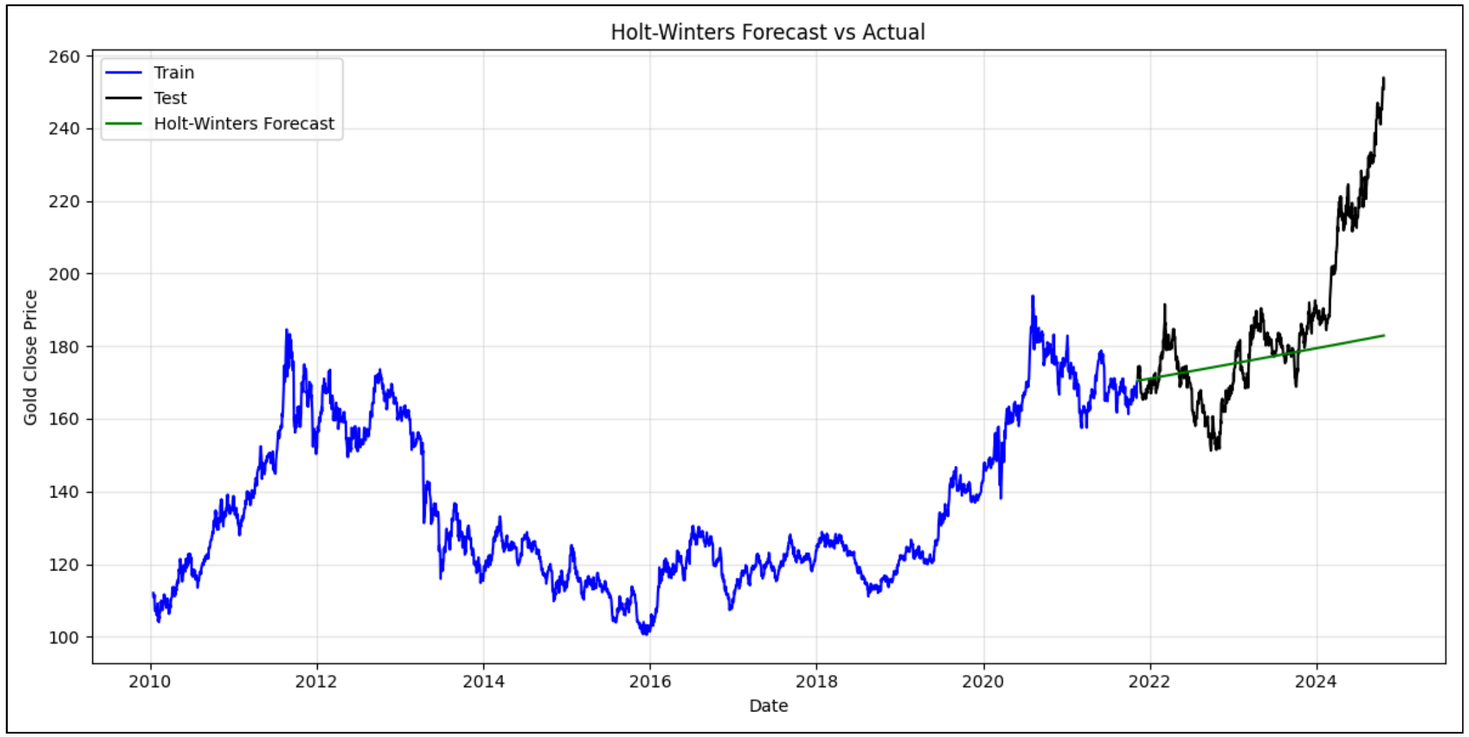
Base Forecasting Models

- Average Model : RMSE = 54.5309, MAE = 49.7513
- Naïve Model : RMSE = 26.7726, MAE = 18.3628
- Drift Model : RMSE = 20.5011, MAE = 13.6719
- SES Model : RMSE = 27.3327, MAE = 18.8494



Holt-Winters Forecasting Model

- Mean Absolute Error (MAE): 14.2446
- Mean Squared Error (MSE): 455.7504
- Root Mean Squared Error (RMSE): 21.3483

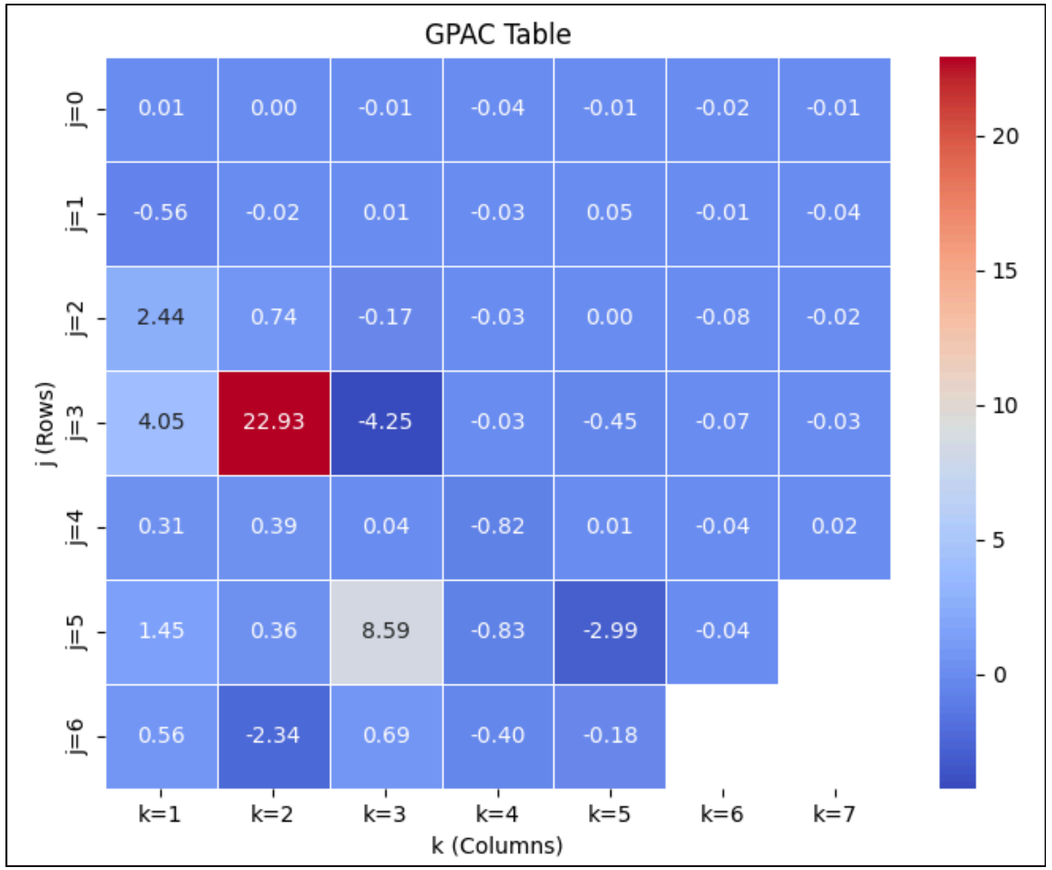


Multiple Linear Regression Model

- Backward elimination
- Good performance, but residuals are autocorrelated

GPAC Table

- GPAC Table does not show any particular order



ARMA Modeling

- Best ARMA(2,2)
- Estimated Parameters: AR[1] = 0.5170, AR[2] = 0.9587, MA[1] = 0.5155, MA[2] = 0.9599
- Residual Q-Test: Q = 36736.3448 → Residuals show autocorrelation

```
Estimated Variance of Error: 1.8939

Confidence intervals:
Param 1: [0.4803, 0.5536]
Param 2: [0.9234, 0.9940]
Param 3: [0.4782, 0.5528]
Param 4: [0.9239, 0.9959]

Standard Deviation of Parameters:
[0.0183 0.0176 0.0186 0.018 ]
```

ARIMA Modeling

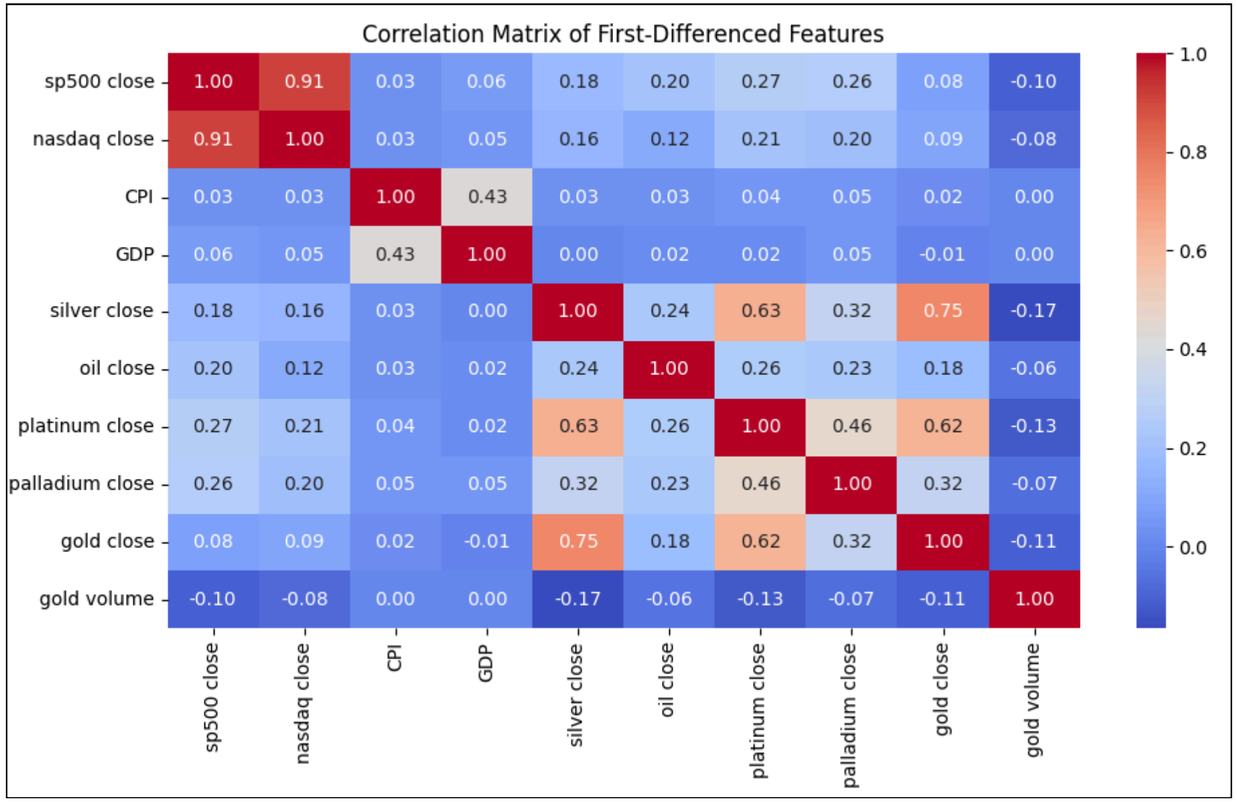
- Best ARIMA(2,1,2)
- Estimated Parameters: [AR]1 = -0.0786, [AR]2 = 0.8756
  MA[1] = 0.0769, MA[2] = -0.8880
- Residual Q-Test: Q = 19.3887 < Q_critical = 62.83 → White

```
Standard Deviations:
SD[ar.L1] = 0.1356
SD[ar.L2] = 0.1379
SD[ma.L1] = 0.1308
SD[ma.L2] = 0.1330
SD[sigma2] = 0.0248

95% Confidence Intervals:
ar.L1: [-0.3443, 0.1872]
ar.L2: [0.6052, 1.1459]
ma.L1: [-0.1794, 0.3332]
ma.L2: [-1.1487, -0.6272]
sigma2: [2.0038, 2.1008]
```

Box-Jenkins Modeling

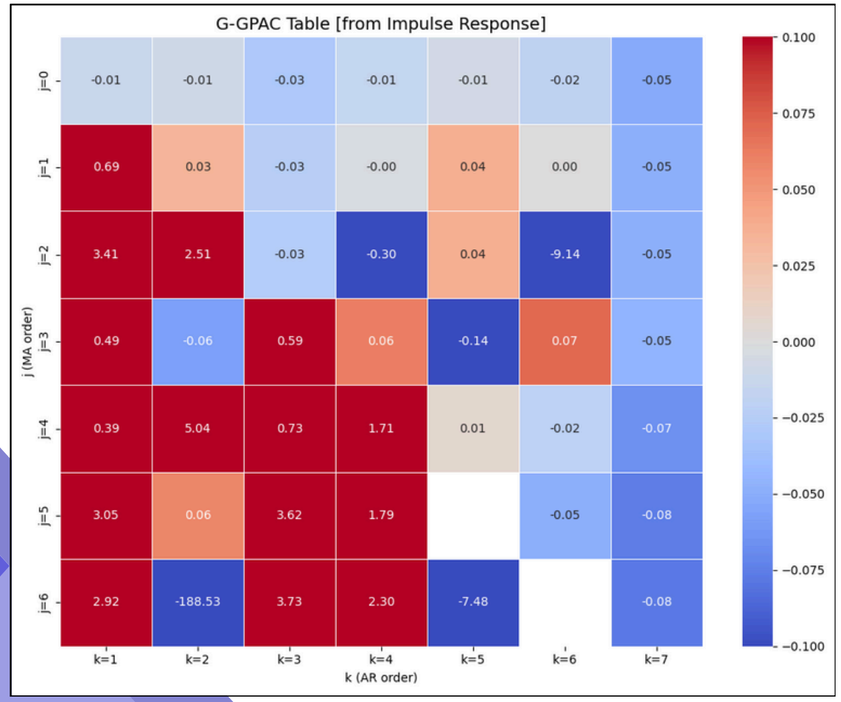
- Input Feature Selection: silver close

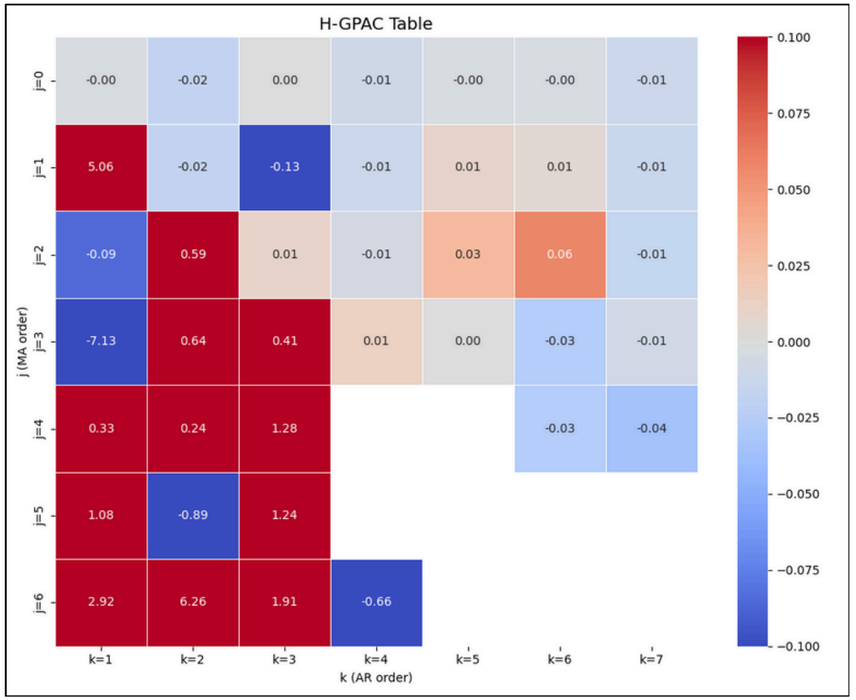


Box-Jenkins Modeling

- Impulse Response, G-GPAC, H-GPAC

```
Shape of Ru(τ) matrix: (51, 51)
First 5 estimated impulse response coefficients:
[ 2.413 -0.035 -0.028 -0.073 -0.031]
```





Box-Jenkins Modeling

- Selected Order: (2,2,2,2) → Filtered Parameters Order: (2,1,2,1)
- Residual diagnostics passed Q and S tests

```
Filtered Significant Parameters (CI excludes 0):
θ[0] = -0.7306 | CI: [-1.1815, -0.2796]
θ[1] = -0.7199 | CI: [-1.1717, -0.2681]
θ[3] = -0.6864 | CI: [-0.7793, -0.5935]
θ[4] = 0.8737 | CI: [0.7815, 0.9659]
θ[5] = 0.6782 | CI: [0.5783, 0.7781]
θ[6] = -0.8603 | CI: [-0.9588, -0.7619]

-- Q-Test (Residual Whiteness) --

--- Q-Test Summary ---
Q-statistic                : 49.1464
Chi-square Critical (α=0.05, dof=42) : 58.1240
Result                     : Residuals are white (Q < Q*)

-- S-Test (Residual-Uncorrelated-With-Input) --
S-stat: 17.2238
Chi-square S* (α=0.05, DOF=17): 27.5871
G(q) is accurate (S < S*)
```

Final Forecast Comparison

- Best: Box-Jenkins (lowest error)

Model	MAE	MSE	RMSE	R^2	Adj R^2	AIC	BIC
ARMA (2,2)	2.470e+80	5.703e+162	2.388e+81	-1.156e+160	-1.162e+160	2.927e+05	2.927e+05
ARIMA (2,1,2)	65.387	4767.097	69.044	-8.666	-8.703	6620.674	6634.656
Box-Jenkins	9.123	135.021	11.620	0.729	0.727	3848.048	3876.019

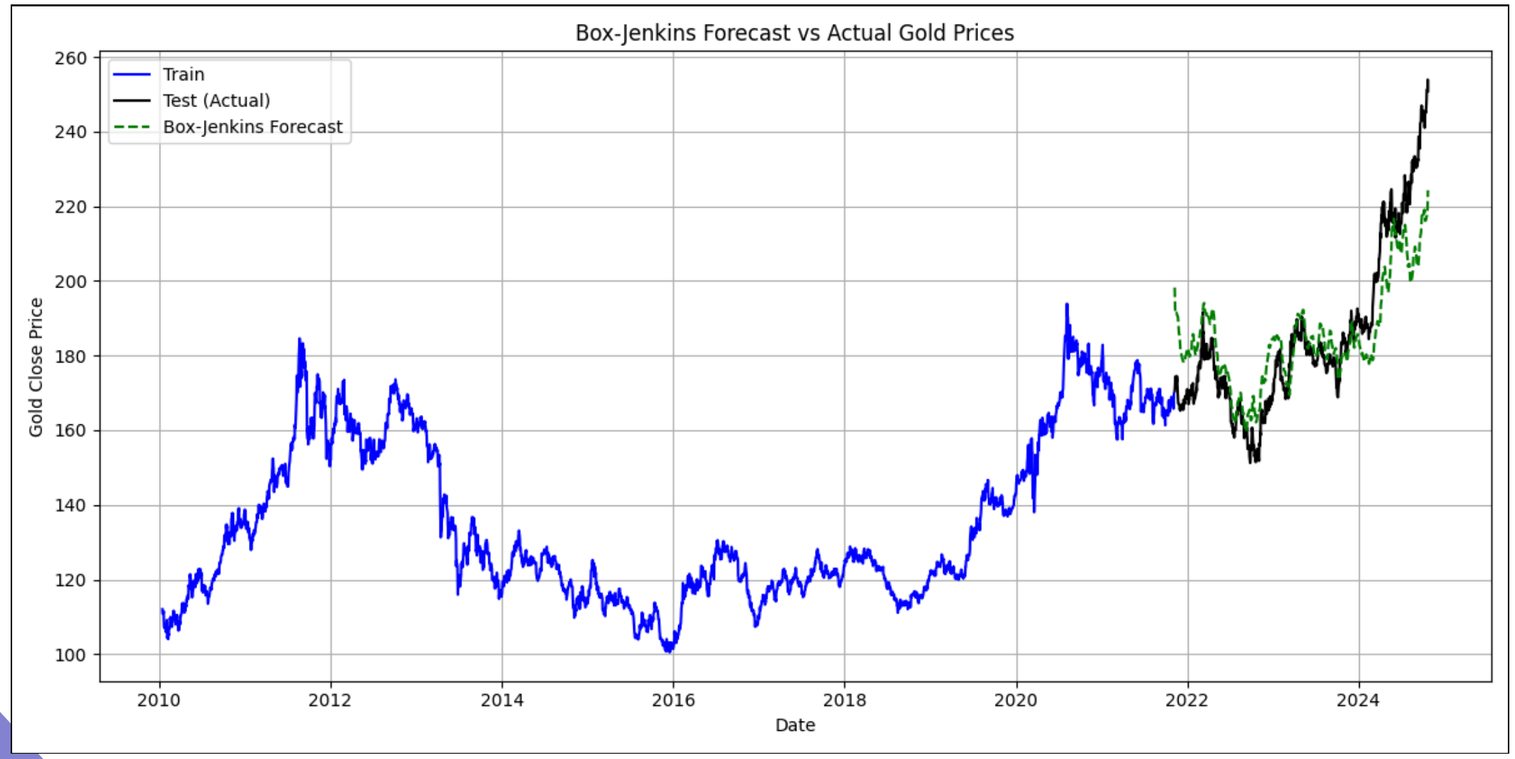
Final Forecast Comparison

Box-Jenkins model (MAE = **9.123**) with other models:

Model	MAE	Improvement Over BJ (%)
Average Model	49.7513	((49.7513 – 9.123) / 49.7513) × 100 ≈ **81.67%**
Naïve Model	18.3628	((18.3628 – 9.123) / 18.3628) × 100 ≈ **50.32%**
Drift Model	13.6719	((13.6719 – 9.123) / 13.6719) × 100 ≈ **33.28%**
SES Model	18.8494	((18.8494 – 9.123) / 18.8494) × 100 ≈ **51.60%**
Holt-Winters	14.2446	((14.2446 – 9.123) / 14.2446) × 100 ≈ **35.94%**
MLR	22.1329	((22.1329 – 9.123) / 22.1329) × 100 ≈ **58.77%**
ARIMA (2,1,2)	65.387	((65.387 – 9.123) / 65.387) × 100 ≈ **86.05%**
ARMA (2,2)	2.470e+80	Invalid (blows up, model failed)

Final Forecast Plot

- Forecast vs Actual for Box-Jenkins
- Forecasts closely follow actual gold price trends
- Short-term fluctuations not always captured
- No major divergence from test values



Conclusion

- Complete modeling pipeline: EDA → Modeling → Forecasting
- Box-Jenkins (2,1,2,1) with silver input delivered best performance
- Passed Q-test & S-test → residuals are white and independent
- Accurate multi-step forecasts using custom-built function

"Gold prices can be reliably forecast using diagnostic-driven time series models — with improved accuracy when leveraging exogenous financial indicators like silver prices."

Thank You

✨ *"From trends to forecasts — mastering gold with models that shine."*